# Girls Build Bots — Junior Automotive Manufacturing

## Badge-aligned workshop guide for robotics-team-led outreach

**Level:** Juniors (Grades 4–5) · **Badge:** Automotive Engineering — Automotive Manufacturing **Series:** Automotive Engineering (Design · Engineering · Manufacturing) **Format:** One 90–120 minute session · **Ratio:** 1 mentor per 4–5 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on paper/craft supplies the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Junior Automotive Engineering badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

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## The through-line: from idea to road

Design it, engineer it, then **manufacture it at scale** — thousands of identical, safe vehicles. Juniors are ready for the real concepts: assembly lines, quality control, bottlenecks, and continuous improvement.

**Vocabulary:** assembly line, station, quality control, defect, bottleneck, throughput, continuous improvement.

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## Automotive Manufacturing (90–120 minutes)

**Badge theme:** How assembly lines, quality control, and teamwork turn one design into many identical vehicles — and how factories keep improving.

### Materials

- A standard origami/paper car design with 6–7 distinct build steps (fold, wheels, windows, stripe, headlights, final check)
- Station signs + supplies pre-sorted per station; a **spec card** at Quality Control showing exactly what a “passing” car looks like
- Two timers; a production tally board (made / passed QC / defects)
- Sticky notes for the improvement round

### Activity 1 — The Scale Problem (10 min)

One great prototype isn’t a car company — factories make *thousands*, identical and safe. How? Assembly lines: each worker masters one step as the product flows past. Today the troop becomes a factory, and we’ll measure it like one.

## Activity 2 — Round 1: Craft Production (Solo Builds) (15 min)

Everyone builds one complete car alone, all steps, 8 minutes. Tally: how many finished? Line them up — how identical are they? Note the honest results on the board (slow, and every car's a little different — that's craft production).

## Activity 3 — Round 2: The Assembly Line (30 min)

Set up stations with one Scout (or pair) per step, plus a **Quality Control inspector** with the spec card who rejects cars missing anything (rejects go back to the failing station — defects are information, not shame). Run 10 minutes. Tally throughput and defects. Almost always: more cars, more consistent — and one station drowning in a pile. Name it: a **bottleneck**.

## Activity 4 — Kaizen Round: Improve the Line (25 min)

Real factories never run the same line twice — they improve it (**continuous improvement**). Team huddle: where was the bottleneck? What would fix it — move a person, split a step, pre-stage supplies? Apply the top ideas and run 10 more minutes. Compare all three rounds on the tally board: solo vs. line vs. improved line. The improvement round almost always wins big — and the *Scouts* engineered that gain.

## Activity 5 — Debrief (10 min) + Wrap (5 min)

Three questions: Why was the line faster? Why did QC matter (would you ride in a car nobody inspected)? What did your improvement change? Count the total fleet and celebrate the team's output — manufacturing is a team sport.

> **Robot moment (optional):** Share for 60 seconds how build season runs like this — parallel stations (wiring, programming, machining), inspections before competition, and constant process improvements between events.

## Completion Checklist — Automotive Manufacturing

By the end of the session, to earn the badge, each Scout can: - [ ] Explain why factories use assembly lines instead of solo craft builds - [ ] Work a station and describe how quality control catches defects - [ ] Identify a bottleneck from production data - [ ] Propose and test an improvement, and compare results across rounds

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# Mentor Guide (for the robotics team)

- **You don't need a robot for this.** Paper cars and timers carry the session; the "robot moment" is a 60-second add-on.
- **Keep the tally board honest and visible.** The numbers make every concept concrete — throughput, defects, improvement.
- **Defects are data.** Coach the QC inspector to reject kindly and specifically ("missing left headlight") — never "you did it wrong."

- **Let the bottleneck hurt a little** in Round 2 before rescuing — the improvement round only shines if the problem was felt.
- **Credit the Scouts' fixes by name** in the debrief ("Maya's pre-staging idea doubled the wheel station") — that's continuous improvement culture.

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## 📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

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*Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Junior Automotive Engineering badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.*